

SDM COLLEGE OF ENGINEERING AND TECHNOLOGY

Dhavalagiri, Dharwad-580002, Karnataka State, India.

Email: cse.sdmcet@gmail.com

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

A Report on WEB TECHNOLOGY PROJECT

COURSE CODE: 22UPCSC405 COURSE TITLE: WEB TECHNOLOGY

SEMESTER: 4TH DIVISION: B

COURSE TEACHER: PROF. INDIRA UMARJI



[Academic Year- 2024-25]

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Submitted By

SI No :	Team Members Name	USN
01	Yukta Y Bhat	2SD23CS128
02	Samanvitha	2SD23CS085

Hobby project :

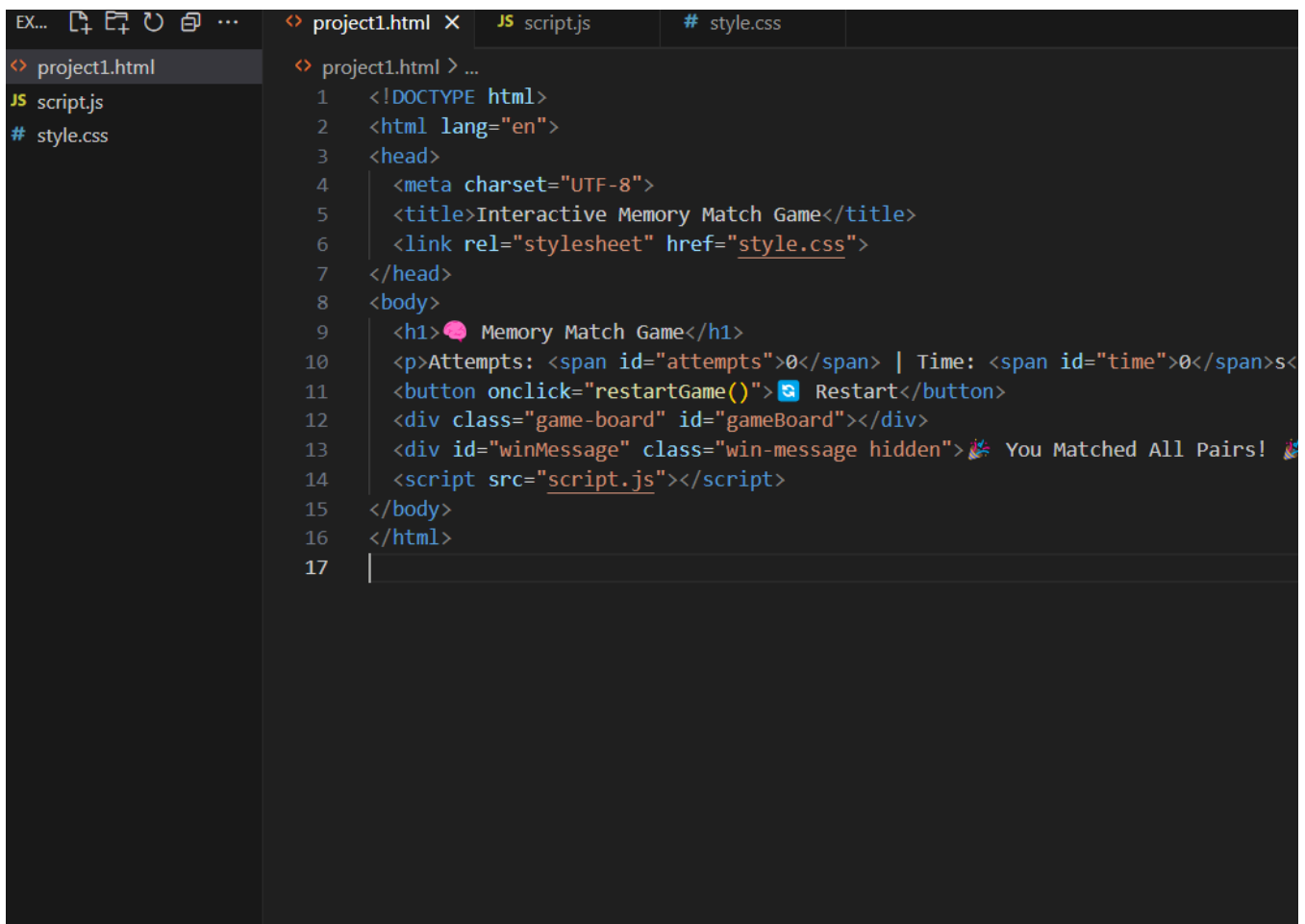
Create a game with a grid of cards, each with a hidden image or symbol. The user clicks on two cards at a time to try and find matching pairs. Keep track of the number of attempts and the time taken.

Core JS Concepts: Arrays, shuffling (algorithm required), event listeners, conditional statements, DOM manipulation (showing/hiding card content, disabling clicks).

HTML File:

It starts with a main heading, "Memory Match Game," followed by a paragraph displaying game statistics: "Attempts:" and "Time:". These likely show the player's current number of attempts and the elapsed time during gameplay. A "Restart" button is included, which, when clicked, triggers the `restartGame()` JavaScript function. Below this, there's a `div` element with the ID `game-board`, intended to hold the memory game cards. Finally, a `div` with the ID `win-message` and class `hidden` (suggesting it's initially hidden) displays "You Matched All Pairs!", which presumably appears upon game completion.

The `script.js` file is linked at the end of the body, indicating that JavaScript handles the game's logic and interactivity.

A screenshot of a code editor with a dark theme. The editor has three tabs at the top: 'project1.html' (active), 'script.js', and 'style.css'. The left sidebar shows a file explorer with 'project1.html', 'script.js', and 'style.css'. The main editor area displays the HTML code for 'project1.html'. The code includes a DOCTYPE declaration, an HTML lang attribute, a head section with a meta charset, a title 'Interactive Memory Match Game', and a link to 'style.css'. The body section contains an h1 heading 'Memory Match Game', a paragraph with 'Attempts:' and 'Time:' placeholders, a 'Restart' button with an 'onclick' event, a 'game-board' div, a 'win-message' div with 'hidden' class and the text 'You Matched All Pairs!', and a script tag linking to 'script.js'. Line numbers 1 through 17 are visible on the left side of the code block.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Interactive Memory Match Game</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Memory Match Game</h1>
  <p>Attempts: <span id="attempts">0</span> | Time: <span id="time">0</span>s</p>
  <button onclick="restartGame()">Restart</button>
  <div class="game-board" id="gameBoard"></div>
  <div id="winMessage" class="win-message hidden">You Matched All Pairs!</div>
  <script src="script.js"></script>
</body>
</html>
```

CSS File:

CSS file named style.css, which is responsible for styling a "Memory Match Game."

The body selector sets basic page styles like font, text alignment, and a light background color. The h1 element has a dark text color.

The .game-board class defines a grid layout for the game cards, arranging them in four columns with specific sizing and spacing. It also includes perspective for 3D transformations.

Individual game cards (.card) are styled with dimensions, a background color, rounded borders, and a pointer cursor. Transitions are applied for smooth visual changes.

The .card.flipped class rotates a card 180 degrees when it's flipped. .card_front and .card_back define the styles for the front and back faces of the cards, including backface-visibility: hidden to hide the back when the card is facing forward.

The .card_front has a specific background and text color, with a 3D rotation, while .card_back has a different background and white text.

The .card.matched class applies a green background color and disables pointer events for cards that have been successfully matched.

Finally, the button element is styled with padding, font size, no border, a specific background color, white text, and rounded corners. The .win-message class styles the victory message with a top margin, font size, and green color, while the .hidden class simply sets display: none to hide elements.

```
EXP... L+ L- U B ... <> project1.html JS script.js # style.css X
<> project1.html
JS script.js
# style.css

# style.css > ...
1  body {
2    font-family: 'Segoe UI', sans-serif;
3    text-align: center;
4    background-color: #f4f6f8;
5  }
6
7  h1 {
8    color: #333;
9  }
10
11  .game-board {
12    display: grid;
13    grid-template-columns: repeat(4, 100px);
14    gap: 15px;
15    justify-content: center;
16    margin-top: 20px;
17    perspective: 1000px;
18  }
19
20  .card {
21    width: 100px;
22    height: 100px;
23    background-color: #3498db;
24    border-radius: 10px;
25    transform-style: preserve-3d;
26    transition: transform 0.6s;
27    position: relative;
28    cursor: pointer;
29  }
30
```

```
EXP... L+ L- U B ... <> project1.html JS script.js # style.css X
<> project1.html
JS script.js
# style.css

# style.css > ...
30
31  .card.flipped {
32    transform: rotateY(180deg);
33  }
34
35  .card .front,
36  .card .back {
37    position: absolute;
38    width: 100%;
39    height: 100%;
40    backface-visibility: hidden;
41    font-size: 2em;
42    display: flex;
43    align-items: center;
44    justify-content: center;
45    border-radius: 10px;
46  }
47
48  .card .front {
49    background-color: #ecf0f1;
50    color: #333;
51    transform: rotateY(180deg);
52  }
53
54  .card .back {
55    background-color: #3498db;
56    color: white;
57  }
```

EXP...    ...

project1.html JS script.js # style.css X

<> project1.html

JS script.js

style.css

style.css > ...

58

.card.matched {

59

background-color: #2ecc71 !important;

60

pointer-events: none;

61

}

62

button {

63

margin-top: 10px;

64

padding: 8px 20px;

65

font-size: 16px;

66

cursor: pointer;

67

border: none;

68

background-color: #2c3e50;

69

color: white;

70

border-radius: 5px;

71

}

72

73

.win-message {

74

margin-top: 20px;

75

font-size: 1.5em;

76

color: green;

77

}

78

79

.hidden {

80

display: none;

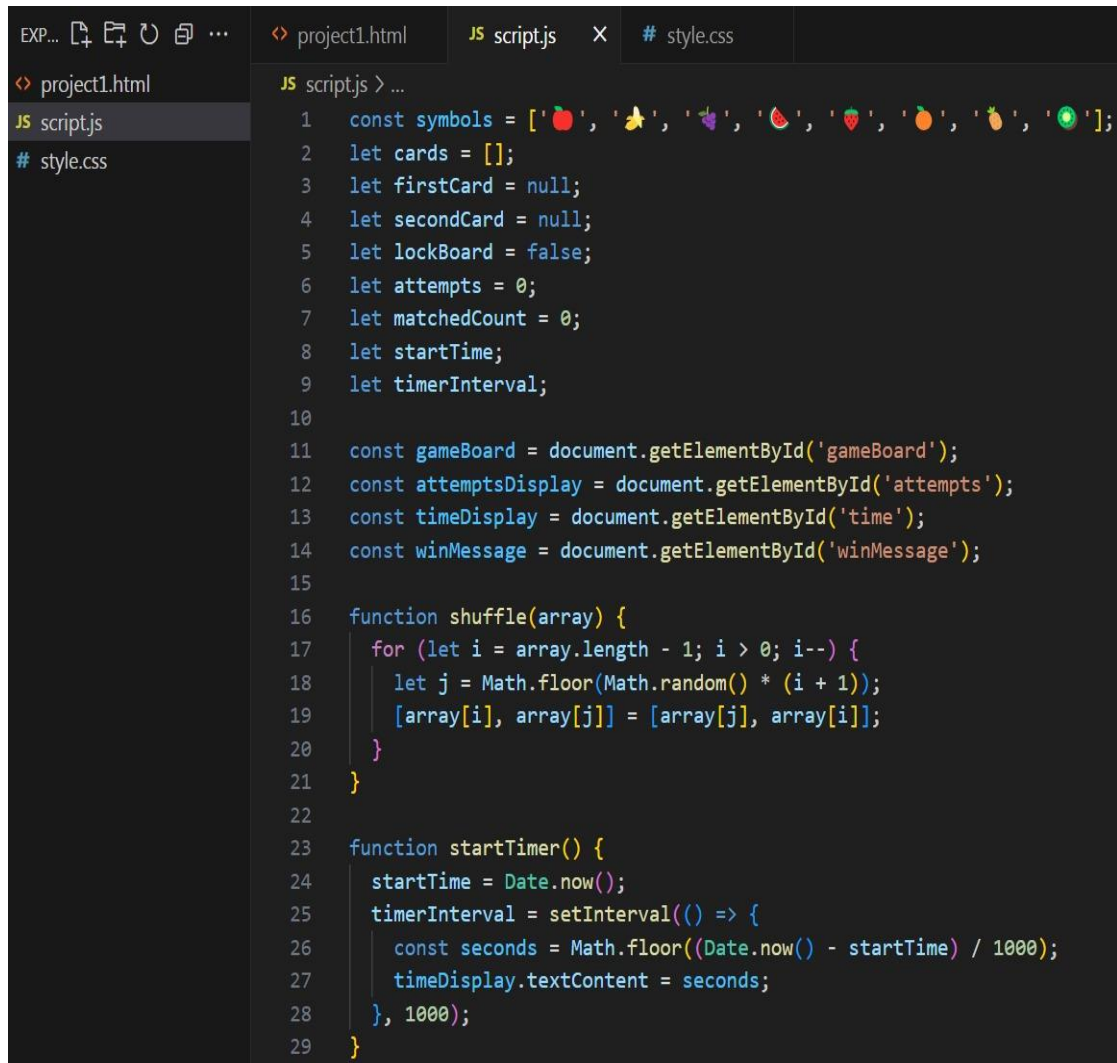
81

}

82

JS File:

This JavaScript code powers a memory-matching card game. It randomly shuffles emoji symbols, creates clickable cards, and tracks user interactions. When two cards are selected, it checks for a match, flips them accordingly, and updates the attempt counter. The game also includes a timer, a restart button, and a win message when all pairs are matched. Core concepts include DOM manipulation, event handling, conditional logic, and interval-based timers.

A screenshot of a code editor with a dark theme. The editor has three tabs at the top: 'project1.html', 'JS script.js', and '# style.css'. The 'JS script.js' tab is active, showing the following JavaScript code:

```
1  const symbols = ['🍎', '🌟', '🍇', '🍉', '🍓', '🍊', '🍌', '🍋'];
2  let cards = [];
3  let firstCard = null;
4  let secondCard = null;
5  let lockBoard = false;
6  let attempts = 0;
7  let matchedCount = 0;
8  let startTime;
9  let timerInterval;
10
11  const gameBoard = document.getElementById('gameBoard');
12  const attemptsDisplay = document.getElementById('attempts');
13  const timeDisplay = document.getElementById('time');
14  const winMessage = document.getElementById('winMessage');
15
16  function shuffle(array) {
17    for (let i = array.length - 1; i > 0; i--) {
18      let j = Math.floor(Math.random() * (i + 1));
19      [array[i], array[j]] = [array[j], array[i]];
20    }
21  }
22
23  function startTimer() {
24    startTime = Date.now();
25    timerInterval = setInterval(() => {
26      const seconds = Math.floor((Date.now() - startTime) / 1000);
27      timeDisplay.textContent = seconds;
28    }, 1000);
29  }
```

```
EXP... [Icons] ... <> project1.html JS script.js X # style.css
<> project1.html JS script.js > ...
JS script.js
# style.css

49 function createBoard() {
50     cards.forEach(symbol => {
51         const card = document.createElement('div');
52         card.classList.add('card');
53         card.dataset.symbol = symbol;
54         const front = document.createElement('div');
55         front.classList.add('front');
56         front.textContent = symbol;
57         const back = document.createElement('div');
58         back.classList.add('back');
59         back.textContent = '?';
60         card.appendChild(front);
61         card.appendChild(back);
62         card.addEventListener('click', handleCardClick);
63         gameBoard.appendChild(card);
64     });
65 }
```

```
EXP... [Icons] ... <> project1.html JS script.js X # style.css
<> project1.html JS script.js > ...
JS script.js
# style.css

30
31 function stopTimer() {
32     clearInterval(timerInterval);
33 }
34
35 function restartGame() {
36     attempts = 0;
37     matchedCount = 0;
38     attemptsDisplay.textContent = 0;
39     timeDisplay.textContent = 0;
40     winMessage.classList.add('hidden');
41     gameBoard.innerHTML = '';
42     cards = [...symbols, ...symbols];
43     shuffle(cards);
44     createBoard();
45     stopTimer();
46     startTimer();
47 }
48
49 function createBoard() {
```



```
EXP...  project1.html  JS script.js  # style.css
project1.html  JS script.js > ...
JS script.js
# style.css

49  function createBoard() {
50      cards.forEach(symbol => {
51          const card = document.createElement('div');
52          card.classList.add('card');
53          card.dataset.symbol = symbol;
54          const front = document.createElement('div');
55          front.classList.add('front');
56          front.textContent = symbol;
57          const back = document.createElement('div');
58          back.classList.add('back');
59          back.textContent = '?';
60          card.appendChild(front);
61          card.appendChild(back);
62          card.addEventListener('click', handleCardClick);
63          gameBoard.appendChild(card);
64      });
65  }
```

```
EX...  project1.html  JS script.js  # style.css
project1.html  JS script.js > ...
JS script.js
# style.css

66
67  function handleCardClick(e) {
68      const clicked = e.currentTarget;
69      if (lockBoard || clicked === firstCard || clicked.classList.contains('flipped')) return;
70
71      clicked.classList.add('flipped');
72
73      if (!firstCard) {
74          firstCard = clicked;
75          return;
76      }
77
78      secondCard = clicked;
79      lockBoard = true;
80      attempts++;
81      attemptsDisplay.textContent = attempts;
82
83      if (firstCard.dataset.symbol === secondCard.dataset.symbol) {
84          firstCard.classList.add('matched');
85          secondCard.classList.add('matched');
86          matchedCount += 2;
87
88          resetTurn();
89
90          if (matchedCount === cards.length) {
91              stopTimer();
92              winMessage.classList.remove('hidden');
93          }
94      } else {
95          setTimeout(() => {
96              firstCard.classList.remove('flipped');
97              secondCard.classList.remove('flipped');
98              resetTurn();
99          }, 1000);
100      }
101  }
```



The image shows a code editor interface with a dark theme. At the top, there are three tabs: 'project1.html', 'JS script.js', and '# style.css'. The 'JS script.js' tab is active. On the left side, there is a file explorer showing the same three files. The main editor area displays the following JavaScript code:

```
102
103  function resetTurn() {
104      [firstCard, secondCard] = [null, null];
105      lockBoard = false;
106  }
107  restartGame();
108
```

view our code on Github : https://github.com/YuktaYBhat/WT_Project1.git

Output:

